Take-Home Exam, Advanced Microeconomics — v2

Problem 1: Overworked Americans – Q1, Q4, Q5, Q7

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1 Q1 – The hours gap and its noisy relation to tax wedges

Question 1. *Document the EU–US (and intra-EU) hours-worked gap in recent years.*

1.1 Definitions

For the working-age (15–64) population: hours per employed worker HE (the intensive object the labour-supply model speaks to), employment rate $e \equiv E/N$, hours per adult $H/N \equiv HE \cdot e$. The Bick–Brüggemann–Fuchs-Schündeln (2019) decomposition $\Delta \log(H/N) = \Delta \log HE + \Delta \log e$ assigns roughly half of the EU–US gap to each margin; we focus on HE.[1]

1.2 The fact

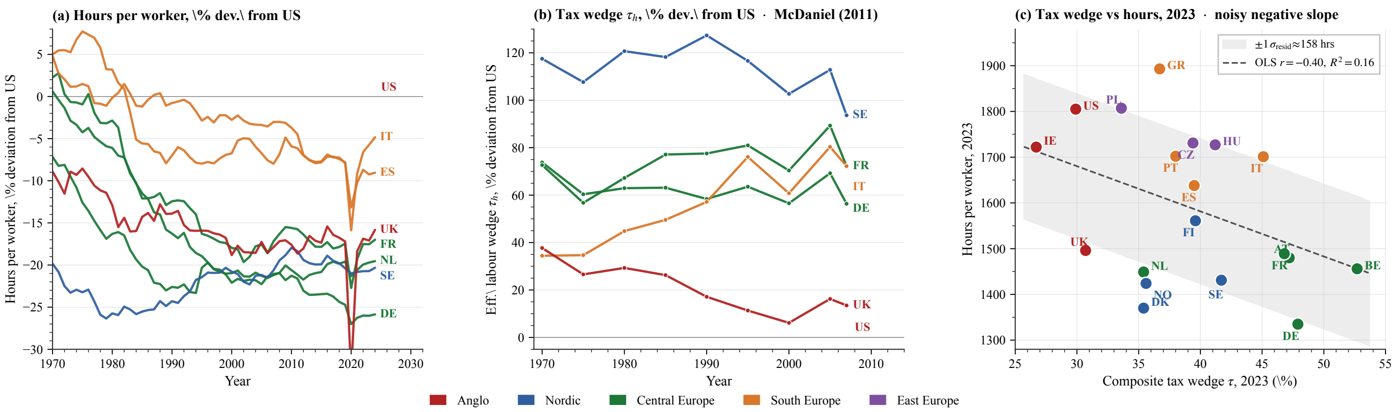

Figure 1: **Three-panel evidence.** (a) Hours per worker, % deviation from US, 1970–2024 (OECD). (b) Effective labour wedge τ_h, % deviation from US, 1970–2007. (c) OECD labour-tax wedge vs hours per worker, 2023 cross-section ($n = 19$); OLS line and $\pm 1\sigma_{\text{resid}}$ band. *Source:* `research/build_figure_panel.py`.

Three readings of Figure 1. **(i)** Hours fell in every focus country relative to the US since 1970 (DE/NL ≈ -25 to -30%; IT/ES ≈ -10 to -15%). **(ii)** The labour-wedge cross-country pattern has no clean structure: Italy's relative wedge drifts up, the UK's narrows, France stays roughly flat — no obvious mirroring of the panel-(a) hours pattern. **(iii)** The 2023 cross-section is signed correctly but noisy: OLS slope -9.9 hrs/pp, $R^2 = 0.16$, residual SD ≈ 158 hours, one third of the entire DE–US gap. Italy ($\tau = 45.1\%$) and Greece (36.7%) work *more* than France (46.8%); the four highest-τ countries (BE, DE, AT, FR) span ≈ 200 hours within a 6 pp wedge band. The regression in panel (c) thus supports two complementary discussions: *(I) identifying the true slope* of the tax–hours relation (OLS signs it correctly but recovers it weakly, with the well-known endogeneity of τ biasing the estimate), and *(II) explaining why observations scatter so widely around that slope* (residual SD ≈ 158 hours — a third of the DE–US gap). Q4 and Q5 develop two institutional channels (an hours cap and union bargaining) that plausibly sit in those residuals as endogenous confounders, not as orthogonal additions to the tax effect.[2]

2 Common framework for Q4 and Q5

2.1 Primitives

We adopt a quadratic-disutility worker, linear consumption, and a linear-quadratic firm-profit function — the simplest closed-form specification that admits both worker- and firm-preferred hours and sign-determinate welfare conditions:

$$V_w(h) = (1-\tau)wh - \tfrac{\alpha}{2}h^2, \;\; h^* = \frac{(1-\tau)w}{\alpha}; \qquad V_f(h) = Ah - \tfrac{\beta}{2}h^2 - wh, \;\; h^{\text{firm}} = \frac{A-w}{\beta}.$$

[1] The extensive margin is shut down throughout this submission — a limitation we acknowledge rather than silently impose. Cross-country e differences are substantial (US 71% vs IT 59% in 2019). The prompt asks about hours, so HE is the relevant object.

[2] Other candidates we do not model: cultural attitudes to work vs. leisure (Alesina, Glaeser & Sacerdote 2005); home-production substitution (Rogerson 2008).

Joint surplus is $W(h) \equiv V_w(h) + V_f(h) = Ah - \tau wh - \frac{\alpha+\beta}{2}h^2$. The wage transfer cancels between worker and firm; only the tax wedge τwh survives. W is concave with maximum

$$h^{\text{soc}}(\tau, \alpha, A, \beta, w) = \frac{A - \tau w}{\alpha + \beta}.$$

The setup is shared by Q4 (regulation) and Q5 (unions). Wages are partial-equilibrium; heterogeneity returns at the end of Q4.

2.2 Baseline calibration and the firm-power ordering

$A = 2,\ w = 1,\ \alpha = \beta = 2,\ \tau = 0.4,\ V_w^0 = 0.05$ gives $h^* = 0.30,\ h^{\text{soc}} = 0.40,\ h^{\text{firm}} = 0.50$. **Justification of the parameter choices.** (i) $\tau = 0.4$ is the population-weighted midpoint between US (0.35) and EU (0.56) labour-tax wedges (OECD *Taxing Wages 2023*, Q1 panel (c)). (ii) The hours scale anchors $h^* = 0.30$ to BLS-OECD hours-per-worker over awake-time at the US point ($\sim$2,000 work hours / $\sim$6,500 awake hours ≈ 0.31, in line with Prescott 2004). (iii) $\alpha = \beta$ is a symmetry assumption that makes the welfare-improving band of (1) collapse to the textbook expression (h^*, h^{firm}), but **only the ordering $h^* < h^{\text{soc}} < h^{\text{firm}}$ is load-bearing** for what follows; the qualitative results are invariant in α, β separately, provided $A > w$, $\tau \in (0,1)$, and the sufficient condition $\alpha(A-w) > \beta(1-\tau)w$ holds (algebraically equivalent to the ordering; verified at calibration: $2 = \alpha(A-w) > \beta(1-\tau)w = 1.2$). (iv) $A = 2$, $w = 1$ are normalisations: the level of A is identified only relative to w, and our results depend on the ratio A/w, not the absolute scale. (v) $V_w^0 = 0.05$ is the worker outside option used to pin the IR-binding wage. It is jointly determined with $w = 1$: at the firm-power baseline IR binds, so $V_w(h^{\text{firm}}; w = 1) = (1-\tau)wh^{\text{firm}} - \frac{\alpha}{2}(h^{\text{firm}})^2 = 0.05$ recovers V_w^0. The discriminant condition $A_v^2 > 4B_vV_w^0$ for a real larger root of (11) is verified in Appendix B.

The ordering says that, when the firm has all the bargaining power, it overshoots the social optimum. We refer to this as the *firm-power baseline* (the firm sets hours on its labour-demand curve at the wage that makes the worker indifferent between working and her outside option). With identical τ and identical preferences, the model predicts no US–EU hours gap under firm power *or* under union bargaining: the gap must come from a country-specific institutional asymmetry, which Q4 (cap) and Q5 (union strength) supply.

3 Q4 – Hours regulation: the optimality condition for a cap

Question 4. *Derive the welfare effects of an hours cap on workers and firms; extend to heterogeneous workers.*

3.1 Modelling choice

A cap $\bar{h}$ is a quantity instrument that takes the firm's labour-demand response as given. *What the trade-off captures.* Workers value leisure (low h^*); firms value output (high h^{firm}); a binding cap forces hours from h^{firm} toward $\bar{h}$. Whether the redistribution is welfare-improving depends on whether $\bar{h}$ lands inside or outside the welfare-improving band defined by joint-surplus concavity: too tight a cap overshoots and destroys joint surplus. *What this buys.* A closed-form welfare condition that depends only on the gap $h^{\text{firm}} - h^{\text{soc}}$, transparent in the heterogeneity extension. *What it costs.* (a) Extensive margin shut down (Q1 footnote): no firm response on number of bodies hired. (b) Wages held at the firm-power IR-binding level (no adjustment of w to absorb the cap). (c) Heterogeneity on α only (workers differ in disutility curvature; we abstract from heterogeneous wages and outside options). The closest sufficient-statistics analogue is Lee–Saez (2012) on the minimum wage, but their welfare logic runs through extensive-margin rationing whereas a binding hours cap rations *intensive*-margin only.

3.2 Setup: firm-power baseline

At $w^* = 1$ pinning the worker's IR at V_w^0, the firm picks $h^{\text{firm}} = (A - w^*)/\beta = 0.50$. Hours sit *above* $h^{\text{soc}} = 0.40$, so the cap has something to correct. With cap $\bar{h}$ the constrained equilibrium is $h^{\text{eq}}(\bar{h}) = \min\{\bar{h},\ h^{\text{firm}}\}$.

3.3 Worker, firm, and joint surplus under a cap

The three surpluses, evaluated at $h^{\text{eq}}(\bar{h})$ relative to the no-cap baseline, are quadratic in $\bar{h}$ around the respective bliss points:

Surplus	Closed form (for $\bar{h} < h^{\text{firm}}$)	Sign
$\Delta V_w(\bar{h}) = V_w(\bar{h}) - V_w(h^{\text{firm}})$	$(\bar{h} - h^{\text{firm}})\big[(1-\tau)w - \frac{\alpha}{2}(\bar{h} + h^{\text{firm}})\big]$	> 0 iff $\bar{h} \in (2h^* - h^{\text{firm}},\ h^{\text{firm}})$
$\Delta V_f(\bar{h}) = V_f(\bar{h}) - V_f(h^{\text{firm}})$	$-\frac{\beta}{2}(h^{\text{firm}} - \bar{h})^2$	< 0 for any $\bar{h} < h^{\text{firm}}$
$\Delta W(\bar{h}) = \Delta V_w + \Delta V_f$	$-\frac{\alpha+\beta}{2}\big[(h^{\text{soc}} - \bar{h})^2 - (h^{\text{soc}} - h^{\text{firm}})^2\big]$	> 0 iff $\bar{h} \in (2h^{\text{soc}} - h^{\text{firm}},\ h^{\text{firm}})$

3.4 Optimality condition (the Q4 result)

Full Lagrangian/FOC derivation of (1), (2), and the aggregate condition is in Appendix A.

Claim. Joint surplus $W(h)$ is strictly concave and quadratic with peak at h^{soc}; hence $W(\bar{h})$ is symmetric in $\bar{h}$ around h^{soc}. The function $\Delta W(\bar{h})$ thus equals zero at $\bar{h} = h^{\text{firm}}$ (no cap) and at the mirror point $2h^{\text{soc}} - h^{\text{firm}}$, and is positive between them. Hence:

$$\Delta W(\bar{h}) \begin{cases} = 0 & \text{if } \bar{h} \geq h^{\text{firm}} \\ > 0 & \text{iff } \bar{h} \in \left(2h^{\text{soc}} - h^{\text{firm}},\, h^{\text{firm}}\right) \\ < 0 & \text{if } \bar{h} < 2h^{\text{soc}} - h^{\text{firm}} \end{cases} \tag{1}$$

provided $h^{\text{firm}} > h^{\text{soc}}$ (which holds at our baseline, $0.50 > 0.40$). The peak gain at $\bar{h} = h^{\text{soc}}$ is $\frac{1}{2}(\alpha + \beta)(h^{\text{firm}} - h^{\text{soc}})^2 = 0.020$ at calibration. At $\alpha = \beta$ the band collapses to $\bar{h} \in (h^*, h^{\text{firm}}) = (0.30, 0.50)$.

Reading. A cap that pulls h *toward* the joint-surplus optimum is unambiguously welfare-improving (the green band of Figure 2c). Two thresholds matter as the cap tightens past h^{soc}. Joint surplus turns negative once $\bar{h} < 2h^{\text{soc}} - h^{\text{firm}} = h^* = 0.30$ (firm losses begin to dominate worker gains). The worker individually keeps benefiting until the cap hits her bliss $h^* = 0.30$; she only becomes worse off than no-cap once the cap tightens further to $\bar{h} < 2h^* - h^{\text{firm}} = 0.10$ — well below the joint-surplus threshold (Appendix A).

3.5 Heterogeneous workers: who is rationed

With worker types indexed by the leisure-disutility curvature α_i (but identical w, τ), the cap forces every type to $\bar{h}$. The individual payoff change $\Delta V_w^i(\bar{h}) = (\bar{h} - h^{\text{firm}})\left[(1-\tau)w - \frac{\alpha_i}{2}(\bar{h} + h^{\text{firm}})\right]$ is positive iff $\alpha_i > \alpha^*(\bar{h})$ where

$$\alpha^*(\bar{h}) = \frac{2(1-\tau)w}{\bar{h} + h^{\text{firm}}}. \tag{2}$$

At $\bar{h} = h^{\text{soc}} = 0.4$: $\alpha^* = 1.333$. **High-α (leisure-loving) workers gain; low-α (income-loving) workers are rationed on the intensive margin**: their preferred hours $h_i^* = (1-\tau)w/\alpha_i$ exceed $\bar{h}$ and the cap forces them below their personal optimum. The cap is therefore a redistributive instrument across worker types: it transfers welfare from low-α workers to high-α workers and away from the firm.

Aggregate condition. Integrating over the type distribution F with mean $\bar{\alpha}$, the cap is welfare-improving on average iff

$$\int \Delta V_w^i(\bar{h})\, dF(\alpha) + \Delta V_f(\bar{h}) > 0.$$

At baseline ($\bar{\alpha} = 2$, $\alpha^* = 1.333$ at $\bar{h} = h^{\text{soc}}$) the cap is welfare-improving on average; a workforce with mean $\bar{\alpha} = 1$ would lose. *Bridge.* A cap helps if the average worker in a country has higher leisure preference than the threshold α^*. If the workforce skews toward income-lovers (α_i low), the cap hurts more workers than it helps and the aggregate verdict flips.

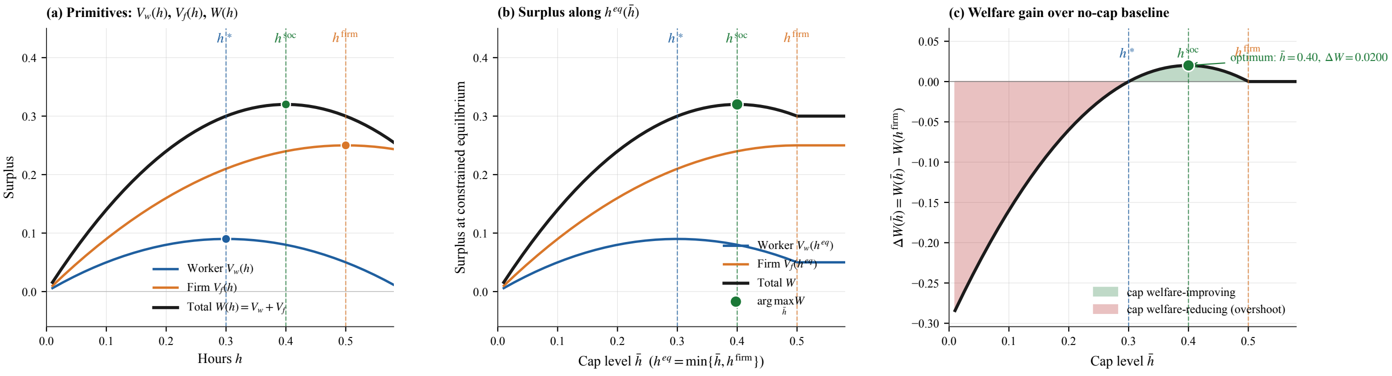

Figure 2: **Hours-cap simulation.** **(a)** Primitives V_w, V_f, W over h with bliss points marked; **(b)** surpluses along $h^{\text{eq}}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}$; **(c)** $\Delta W(\bar{h})$ vs no-cap baseline — green band is the welfare-improving region of (1), red is overshoot. Maximum gain $\Delta W = 0.020$ at $\bar{h} = h^{\text{soc}} = 0.40$. *Source:* `research/sim_mechanisms.py`.

Empirical anchor. The 35-hour Aubry reform (France 1998–2000) documented by Estevão–Sá (2008) found $\approx 10 - 13\%$ hours cut and near-zero employment effect, consistent with a binding cap at $\bar{h}/h^{\text{firm}} \approx 0.875$ inside the welfare-improving band of (1).[3]

[3] The near-zero employment effect is consistent with our model only because we have shut down the extensive margin (see the Q1 footnote): if a binding cap reduces hours per worker, firms may substitute workers for hours, an effect the representative-agent setup rules out by construction. Sectoral composition (a large public sector with multi-year collective contracts) is also outside the model.

4 Q5 – A single union as welfare-restorer in a two-sector economy

Question 5. *Show how sectoral productivity shifts affect hours under competition vs. unions; what data would test the hypotheses?*

4.1 Modelling choice

What the trade-off captures. The Walrasian benchmark rewards productivity through wages, so a sectoral shock is absorbed by reallocating hours; the union changes the channel of absorption (wages take more, hours take less) and, under downward wage rigidity, makes that channel asymmetric across positive and negative shocks. *What this buys.* A single bargain primitive (η) delivers a continuous comparative static between the firm-power baseline ($\eta = 0$) and the worker-power limit ($\eta \to 1$), nesting the Q4 distortion and the Walrasian benchmark as endpoints. *What it costs.* (a) A single encompassing union, following Calmfors–Driffill (1988): encompassing (national/centralised) unions internalise their macroeconomic effect, so their objective aligns with worker surplus rather than with subgroup rent extraction, letting us match the European centralised-bargaining institutional reality. (b) Right-to-manage instead of efficient bargaining — the standard tractability convention for unionised wage-setting. (c) Informal labour and the public sector are outside the model.[4]

4.2 Two-sector economy: competitive benchmark

Two sectors $j \in \{\text{G}, \text{B}\}$ (good, bad) with productivities $A_{\text{G}} > A_{\text{B}}$, drawn from a mean-preserving sectoral shock around a symmetric pre-shock benchmark $A_{\text{pre}} = 2$ (we set $A_{\text{G}} = 2.2$, $A_{\text{B}} = 1.8$, a $\pm 10\%$ shock chosen as a conservative magnitude for developed-world cross-sectoral TFP dispersion; Hsieh–Klenow (2009) document substantially wider dispersion in developing economies, so $\pm 10\%$ sits below their headline range). Within each sector a representative worker–firm pair has the quadratic primitives of §3.

Competitive (Walrasian) benchmark. With perfect labour mobility across sectors, sector wages clear at the marginal value product: the firm's labour demand is $h^d_j(w) = (A_j - w)/\beta$, and the worker's IR pins w^{FM}_j such that hours equal $h^{\text{firm}}_j(w^{\text{FM}}_j)$. The headline competitive comparative static is:

$$\frac{\partial h^{\text{FM}}_j}{\partial A_j} > 0, \qquad \frac{\partial w^{\text{FM}}_j}{\partial A_j} > 0.$$

Hours follow productivity: positive shocks raise hours and wages in the good sector; negative shocks lower them in the bad sector. Cross-sectoral allocation is efficient given the shock.

4.3 Two-sector economy: unionised version

A single union covers both sectors. Per sector, union and firm right-to-manage Nash-bargain over w_j with worker weight $\eta \in (0, 1)$:

$$w_j(\eta; A_j) = \arg\max_{w} \left[V_w(w, h^d_j(w)) - V^0_w\right]^{\eta} V_f(w, h^d_j(w); A_j)^{1-\eta}, \qquad h^d_j(w) = (A_j - w)/\beta. \tag{3}$$

The first-order condition reduces to a quadratic in $x_j \equiv A_j - w_j$ (Appendix B) with limits $h_j(0) = h^{\text{firm}}_j$ and $h_j(1) < h^*(w_j(1))$; $\partial w_j/\partial\eta > 0$, $\partial h_j/\partial\eta < 0$ throughout.

Welfare-restorer. Under the firm-power baseline $h^{\text{firm}}_j > h^{\text{soc}}_j > h^*_j$, so as η rises the union pulls hours *toward* the social optimum: joint surplus rises while $h_j(\eta) \in (h^{\text{soc}}_j, h^{\text{firm}}_j)$ and falls thereafter — the overshoot logic of Q4. Worker surplus is monotone in η, so any planner with positive worker weight strictly prefers $\eta > 0$.

US vs. EU calibration. OECD ICTWSS coverage maps US ($\sim 12\%$) to $\eta^{US} \approx 0.15$ and population-weighted EU coverage ($\sim$60–70% across DE/FR/IT/SE under *erga omnes*) to $\eta^{EU} \approx 0.5$. At these calibrated values we get $h^{US} \approx 0.46$ at $\eta^{US} = 0.15$ and EU mean hours $\frac{1}{2}(h_{\text{G}} + h_{\text{B}}) \approx 0.38$ at $\eta^{EU} = 0.5$ (sectoral $h_{\text{G}} \approx 0.42$, $h_{\text{B}} \approx 0.33$), an EU–US gap of order ≈ 0.08 ($\approx 19\%$ in logs) — the right order of magnitude to absorb a substantial fraction of the gap documented in Q1.

4.4 Sectoral shocks: union pass-through and asymmetry

Substituting $A_{\text{G}}, A_{\text{B}}$ into (3) at $\eta = 0.5$ (Figure 3): $w^U_{\text{G}} = 1.354$, $h^U_{\text{G}} = 0.423$ and $w^U_{\text{B}} = 1.132$, $h^U_{\text{B}} = 0.334$. The good sector splits the productivity gain into wages ($\Delta w/w \approx +25\%$) and reduced hours ($\Delta h/h \approx -24\%$); joint surplus falls only $\approx 5\%$ ($W^{\text{FM}}_g = 0.363 \to W^U_g = 0.344$), with the bulk of the redistribution being a wage-rent transfer rather than a deadweight loss.

Asymmetry under downward wage rigidity. If unions refuse to renegotiate wages downward, the bad-sector wage is stuck at $w_{\text{pre}} = 1.242$ and hours collapse to $(A_{\text{B}} - w_{\text{pre}})/\beta = 0.279$ (36% below free market;

[4] An alternative model would have the union charge a share s of worker income to finance itself; the union's revenue then scales with worker income, so its incentive maps directly onto worker surplus and the alignment result below is unchanged.

firm profit 60% below). Headline: **good shocks pass through diluted (absorbed in wage rents); bad shocks pass through amplified (rigidity forces hours to absorb the entire shock)**.

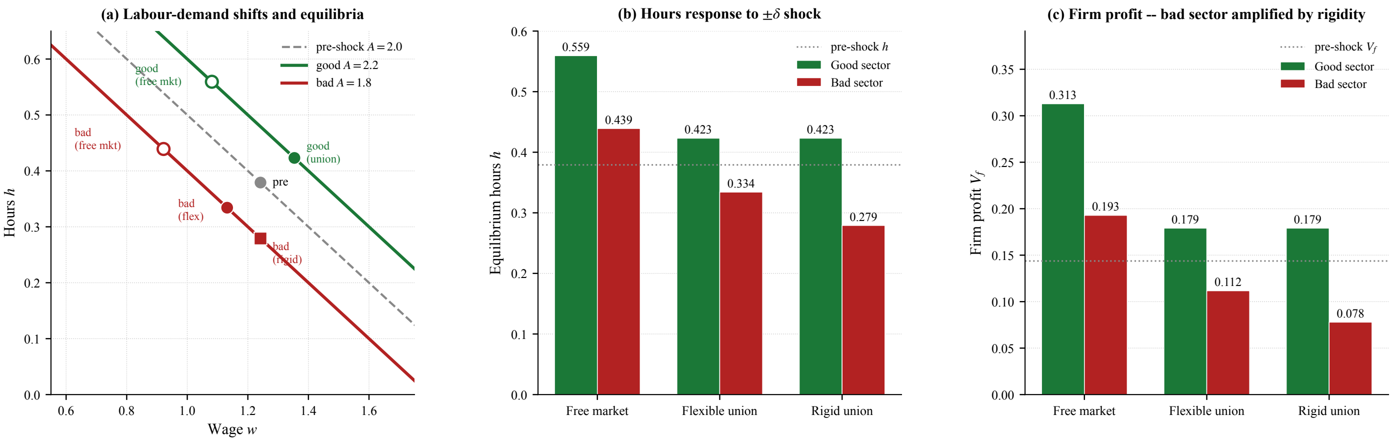

Figure 3: **Asymmetric pass-through under three regimes. (a)** Labour-demand shifts and equilibria: rigid bad-sector equilibrium sits on the new demand at the stuck pre-shock wage. **(b)** Hours response: good-sector hours rise under all regimes; bad-sector rigid is 36% below free market. **(c)** Firm profit: good-sector profit rises under all regimes; bad-sector rigid is 60% below free market. *Source:* `research/sim_models_v2.py:make_shocks`.

4.5 Data to test the hypotheses

Three falsifiable predictions: (a) hours decrease in union bargaining strength conditional on productivity; (b) the US–EU gap rises with the coverage gap; (c) bad-sector shocks generate larger hours responses in unionised economies. Required data: sectoral hours panels (OECD STAN, EU KLEMS), coverage by sector (ICTWSS), collective-agreement wage indices. Identification of the rigidity asymmetry exploits the post-2000 manufacturing decline in DE and IT, where sectoral hours and employment fell sharply but real wages did not.

5 Q7 – Welfare comparison: a revealed-preference reading

Question 7. *Write a stylised two-system model and derive conditions under which the European system is welfare-enhancing relative to the US.*

Methodological note. Each of Q4 and Q5 admits a much deeper treatment than we provide — full GE, dynamic bargaining, sectoral migration, heterogeneous firms. We deliberately treat both shallowly because the prompt's substantive question is comparative (Q7: "derive conditions under which the European system is welfare-enhancing"), and a comparative answer requires coverage of the institutional menu, not depth in any one channel. Q4 and Q5 each show that their respective channel can absorb the residual the Prescott mechanism leaves unexplained; Q7 is where these channels are confronted with the US–EU mapping and the welfare verdict is rendered.

5.1 Modelling choice

The Q4 cap and the Q5 union, in our specification, are both quantity instruments that pull hours from h^{firm} toward h^{soc}.[5] Whether a given configuration is welfare-enhancing depends on *whose* welfare we are summing. The natural representation of the cross-country welfare disagreement is the Saez–Stantcheva (2016) generalised welfare weight: the planner attaches premium $\lambda \in [0, 1]$ to a worker dollar over a firm-owner dollar,

$$W^{\lambda}(\eta) = (1 + \lambda)\, V_w(\eta) + (1 - \lambda)\, V_f(\eta).$$

$\lambda = 0$ is equal-weight utilitarianism, $\lambda = 1$ is Rawlsian. Following Alesina–Angeletos (2005), λ is a country-specific primitive: it captures fairness beliefs (whether income inequality is seen as deserved or as luck), which can differ across countries even at common preferences and technology. We compare the two systems through the lens of this single parameter.

[5] We focus on hours throughout. The wage channel is implicit: the union's effect on h operates through a bargained wage premium, and the Saez–Stantcheva λ assigns welfare value to the firm-to-worker transfer that this premium effects. We do not separately analyse cross-country wage differences (Krueger–Summers 1988), which the prompt does not ask about and our partial-equilibrium calibration cannot speak to.

5.2 Welfare-optimum bargaining weight: closed form

Maximising $W^{\lambda}(\eta)$ along the bargained path of (3) gives the welfare-optimum reduced-form variable $x^* \equiv A - w^* = \beta h^*$:

$$x^*(\lambda) \;=\; \frac{0.3\,A\,(1+\lambda)}{0.6+1.6\,\lambda} \qquad \text{(at baseline; derivation in } \texttt{sim\_models\_v2.py}\text{)}.$$

Substituting $x^*(\lambda)$ into the Nash-bargain quadratic recovers the *welfare-optimum bargaining weight*:

$$\eta^*(\lambda) = \frac{1.2\,x^* - 1.1\,(x^*)^2 - 0.1}{0.6\,x^* - 0.1}.$$

η^* is monotonically increasing in λ, with $\eta^*(0)=0$ (firm-power optimal under equal weights) and $\eta^*(1)=1$ (full union under Rawlsian).

5.3 Confronting data: observed unionisation in US and EU

We map the Nash-bargain weight η to OECD bargaining-coverage data: $\hat{\eta}^{US} \approx 0.15$ (coverage $\sim$ 12%), $\hat{\eta}^{EU} \approx 0.50$ (coverage $\sim$ 60% population-weighted across DE/FR/IT/SE).[6] These two observations admit two competing readings.

Reading A — common λ. If both countries share the HSV-anchored $\lambda = 0.30$, the welfare-optimum is $\eta^*(0.30)=0.58$, so the US ($\hat{\eta}=0.15$) is heavily under-unionised and the EU ($\hat{\eta}=0.50$) only mildly. The implied losses $W^{0.30}(\eta^*) - W^{0.30}(\hat{\eta})$ are 0.022 for the US and 0.001 for the EU — the US leaves $\sim$8% of joint surplus on the table. We flag this reading but do not endorse it (see Reading B).

Reading B — country-specific λ. Inverting $\eta^*(\lambda)$ on observed coverage:

$$\hat{\eta}^{US}=0.15 \implies \lambda^{US,\text{rev}}\approx 0.05, \qquad \hat{\eta}^{EU}=0.50 \implies \lambda^{EU,\text{rev}}\approx 0.24.$$

Each country's institutions are then *at-optimum given its own fairness primitive*; cross-country fairness-belief gap is $\Delta\lambda \approx 0.19$. The hours drag from the union channel is small in the US ($h^{\text{firm}} \to 0.45$, -9%) and larger in the EU ($h^{\text{firm}} \to 0.38$, -24%). **This is the missing variable in Q1 panel (c)**: the cross-country hours variation that the noisy tax–hours relation cannot pin down is reconstructed by the λ-driven union channel.

5.4 Verdict

The two readings give compatible verdicts on *whether* the EU system is more welfare-enhancing, but disagree on *why*.

- *Reading A* (common $\lambda = 0.30$): **EU is unconditionally welfare-better**, because EU institutions are closer to any reasonable planner's optimum than US institutions are. The mechanism is institutional implementation, not preferences.
- *Reading B* (country-specific λ): **neither country is welfare-better in absolute terms** — each implements its own optimum. The cross-country hours gap reflects a fairness-belief gap, which is the missing variable in the noisy Q1 cross-section.

The two readings are *empirically distinguishable* (Figure 4): Reading A predicts all countries on the same horizontal line $\eta = \eta^*(0.30)$, which the observed range (US 0.15, FR 0.98, SE 0.88, DE 0.52) clearly violates. **The data prefer Reading B** — but Reading A is the cleaner welfare argument.[7]

[6] The mapping is a coarse linear proxy: it overstates Nordic high-density regimes and understates French *erga omnes* extension regimes, where coverage is high but bargaining power is weak.

[7] The cap channel (Q4) is orthogonal at our calibration: the union alone pulls h below any sensible cap, so Q4 is a counterfactual backstop, not an active driver. Outside the labour-market block, dominant US–EU welfare differences run through non-labour channels (life expectancy, consumption levels — Jones–Klenow 2016); our verdict is sign-determinate but quantitatively second-order.

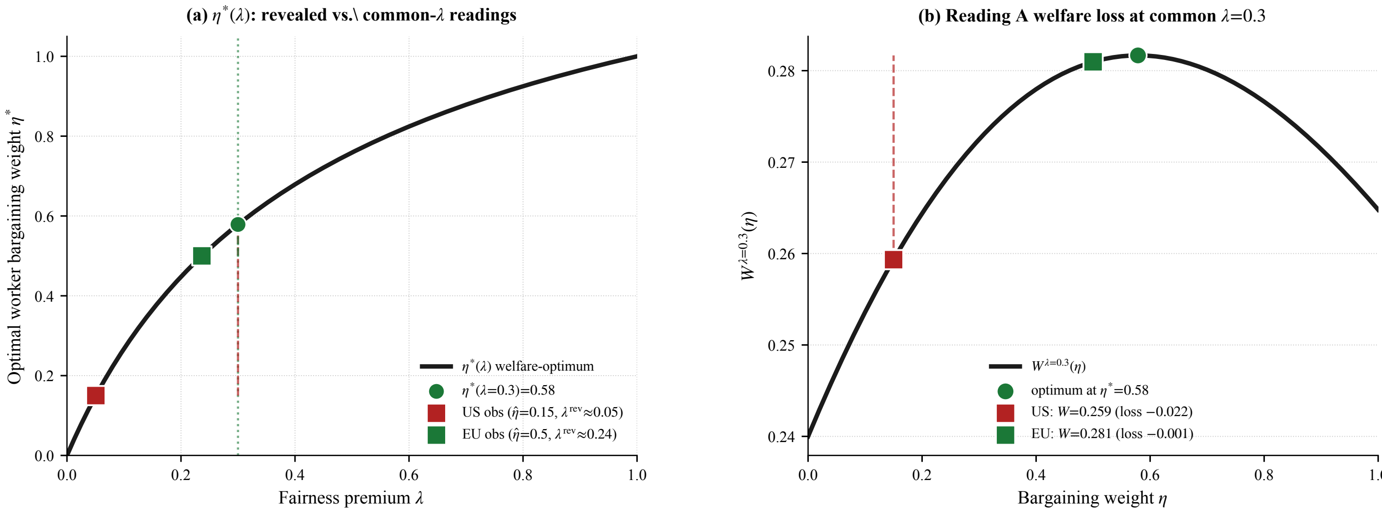

Figure 4: **Q7 revealed-preference diagnostic.** **(a)** The $\eta^*(\lambda)$ welfare-optimum locus (black). Observed US and EU points sit on the locus at their revealed-preference λ^{rev} (Reading B). Vertical dashed segments show the deviation under common $\lambda = 0.30$ (Reading A): both countries below the optimum, US far more so. **(b)** $W^{\lambda=0.30}(\eta)$: peak at $\eta^* = 0.58$. US loss -0.022 (under-unionised by 0.43); EU loss -0.001 (essentially at-optimum). *Source:* `sim_models_v2.py:make_q7_eta_lambda`.

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A Formal derivation of the Q4 cap results

This appendix derives the cap surpluses, the welfare band (1), the heterogeneous-worker threshold (2), and the aggregate condition from primitives via Lagrangian and FOC.

A.1 Primitives, FOCs, and the three bliss points

The worker and firm payoffs are

$$V_w(h) = (1-\tau)wh - \tfrac{\alpha}{2}h^2, \qquad V_f(h) = Ah - \tfrac{\beta}{2}h^2 - wh.$$

Worker FOC. $V_w'(h) = (1-\tau)w - \alpha h = 0 \implies h^* = (1-\tau)w/\alpha$ (the worker's preferred hours absent any constraint).

Firm FOC. $V_f'(h) = A - \beta h - w = 0 \implies h^{\text{firm}} = (A-w)/\beta$ (the firm's unconstrained labour demand at wage w).

Joint-surplus FOC. Adding the two payoffs the wage transfer cancels and

$$W(h) = V_w(h) + V_f(h) = Ah - \tau wh - \tfrac{\alpha+\beta}{2}h^2, \qquad W'(h) = A - \tau w - (\alpha+\beta)h = 0 \implies h^{\text{soc}} = \frac{A - \tau w}{\alpha + \beta}.$$

At the calibration $(A, w, \tau, \alpha, \beta) = (2, 1, 0.4, 2, 2)$: $h^* = 0.30$, $h^{\mathrm{soc}} = 0.40$, $h^{\mathrm{firm}} = 0.50$, so the firm-power ordering $h^* < h^{\mathrm{soc}} < h^{\mathrm{firm}}$ holds.

A.2 Firm's constrained programme under a cap

In the firm-power baseline the firm chooses h to maximise its own surplus subject to (i) the regulatory cap $h \le \bar{h}$ and (ii) the worker's individual-rationality (IR) constraint $V_w(h) \ge V_w^0$, with the wage chosen so that IR binds.

Why IR enters the programme. Without IR the firm-power problem is degenerate (the firm could set w arbitrarily low and extract unbounded rents); IR pins the wage at the lowest level that satisfies the worker's outside option V_w^0, closing the model.

The Lagrangian is

$$\mathcal{L}(h, \mu, \lambda) = V_f(h) - \mu(h - \bar{h}) + \lambda\big(V_w(h) - V_w^0\big), \qquad \mu, \lambda \ge 0.$$

The FOC and KKT slackness conditions read

$$V_f'(h) + \lambda V_w'(h) = \mu, \qquad \mu(h - \bar{h}) = 0, \qquad \lambda(V_w(h) - V_w^0) = 0.$$

The wage is set so that IR binds, which fixes λ; the inner choice of h then collapses to $\max_h V_f(h)$ s.t. $h \le \bar{h}$. Two cases:

- **Cap slack** ($\bar{h} \ge h^{\mathrm{firm}}$): $\mu = 0$, $h^{\mathrm{eq}} = h^{\mathrm{firm}}$.
- **Cap binding** ($\bar{h} < h^{\mathrm{firm}}$): $h^{\mathrm{eq}} = \bar{h}$ and $\mu = V_f'(\bar{h}) = A - w - \beta\bar{h} > 0$ (the shadow price of the cap is the firm's marginal value of an extra hour at the constrained equilibrium).

Combined: $h^{\mathrm{eq}}(\bar{h}) = \min\{\bar{h},\, h^{\mathrm{firm}}\}$. All subsequent surpluses are evaluated at this h^{eq} relative to the no-cap baseline h^{firm}.

A.3 Quadratic identities for the three surpluses

Any concave quadratic $g(h)$ with bliss point $h^\dagger$ and curvature $-\kappa$ satisfies

$$g(h) - g(h^\dagger) = -\tfrac{\kappa}{2}(h - h^\dagger)^2. \tag{4}$$

Apply (4) to V_f (bliss h^{firm}, curvature β) and to W (bliss h^{soc}, curvature $\alpha + \beta$):

$$\Delta V_f(\bar{h}) \equiv V_f(\bar{h}) - V_f(h^{\mathrm{firm}}) = -\tfrac{\beta}{2}\big(h^{\mathrm{firm}} - \bar{h}\big)^2 \le 0, \tag{5}$$

$$\begin{aligned}\Delta W(\bar{h}) &\equiv W(\bar{h}) - W(h^{\mathrm{firm}}) = \big[W(\bar{h}) - W(h^{\mathrm{soc}})\big] - \big[W(h^{\mathrm{firm}}) - W(h^{\mathrm{soc}})\big] \\ &= -\tfrac{\alpha+\beta}{2}\Big[(h^{\mathrm{soc}} - \bar{h})^2 - (h^{\mathrm{soc}} - h^{\mathrm{firm}})^2\Big].\end{aligned} \tag{6}$$

For ΔV_w the reference point h^{firm} is *not* the worker's bliss, so we expand directly:

$$\begin{aligned}\Delta V_w(\bar{h}) &= V_w(\bar{h}) - V_w(h^{\mathrm{firm}}) \\ &= (1-\tau)w(\bar{h} - h^{\mathrm{firm}}) - \tfrac{\alpha}{2}\big(\bar{h}^2 - (h^{\mathrm{firm}})^2\big) \\ &= (\bar{h} - h^{\mathrm{firm}})\Big[(1-\tau)w - \tfrac{\alpha}{2}(\bar{h} + h^{\mathrm{firm}})\Big],\end{aligned} \tag{7}$$

using the factorisation $\bar{h}^2 - (h^{\mathrm{firm}})^2 = (\bar{h} - h^{\mathrm{firm}})(\bar{h} + h^{\mathrm{firm}})$. (5), (6), (7) are precisely the three rows of the table in §3.3.

A.4 Welfare-improving band: derivation of (1)

For $\bar{h} < h^{\mathrm{firm}}$, (6) gives

$$\Delta W(\bar{h}) > 0 \iff (h^{\mathrm{soc}} - \bar{h})^2 < (h^{\mathrm{soc}} - h^{\mathrm{firm}})^2 \iff |h^{\mathrm{soc}} - \bar{h}| < h^{\mathrm{firm}} - h^{\mathrm{soc}},$$

using $h^{\mathrm{firm}} > h^{\mathrm{soc}}$. Removing absolute values:

$$-(h^{\mathrm{firm}} - h^{\mathrm{soc}}) < h^{\mathrm{soc}} - \bar{h} < h^{\mathrm{firm}} - h^{\mathrm{soc}} \iff 2h^{\mathrm{soc}} - h^{\mathrm{firm}} < \bar{h} < h^{\mathrm{firm}},$$

which is (1). Maximising (6) in $\bar{h}$ gives the peak $\bar{h} = h^{\mathrm{soc}}$ with gain

$$\Delta W(h^{\mathrm{soc}}) = \tfrac{\alpha+\beta}{2}\big(h^{\mathrm{firm}} - h^{\mathrm{soc}}\big)^2 = \tfrac{4}{2}(0.10)^2 = 0.020.$$

Band collapse at $\alpha = \beta$. Substituting the bliss-point formulas,

$$2h^{\mathrm{soc}} - h^{\mathrm{firm}} = \frac{2(A - \tau w)}{\alpha + \beta} - \frac{A - w}{\beta} \xrightarrow{\alpha=\beta} \frac{A - \tau w}{\beta} - \frac{A - w}{\beta} = \frac{(1-\tau)w}{\beta} = h^*.$$

Hence at $\alpha = \beta$ the lower band-edge equals the worker's bliss h^*, and (1) simplifies to $\bar{h} \in (h^*, h^{\mathrm{firm}}) = (0.30, 0.50)$.

A.5 Worker-surplus band and the two thresholds

The sign of ΔV_w in row 1 of the §3.3 table comes from (7). For $\bar{h} < h^{\text{firm}}$, the leading factor $(\bar{h} - h^{\text{firm}}) < 0$, so $\Delta V_w(\bar{h}) > 0$ iff the bracket is also negative:

$$(1-\tau)w < \tfrac{\alpha}{2}(\bar{h} + h^{\text{firm}}) \iff \bar{h} + h^{\text{firm}} > \frac{2(1-\tau)w}{\alpha} = 2h^* \iff \bar{h} > 2h^* - h^{\text{firm}}.$$

Combined with $\bar{h} < h^{\text{firm}}$ (cap binding), the worker-surplus band is

$$\Delta V_w(\bar{h}) > 0 \iff \bar{h} \in \big(2h^* - h^{\text{firm}},\, h^{\text{firm}}\big). \tag{8}$$

The peak of ΔV_w is at the worker's bliss $\bar{h} = h^*$, with magnitude

$$\Delta V_w(h^*) = V_w(h^*) - V_w(h^{\text{firm}}) = \tfrac{\alpha}{2}(h^{\text{firm}} - h^*)^2 = \tfrac{2}{2}(0.20)^2 = 0.04$$

at calibration. Outside the band ($\bar{h} < 2h^* - h^{\text{firm}}$), the cap is so tight that the worker is forced past her bliss far enough to be worse off than at h^{firm}.

The two thresholds reconciled. (1) (joint-surplus band) and (8) (worker-surplus band) have different lower edges:

$$\underbrace{2h^{\text{soc}} - h^{\text{firm}}}_{\text{joint turns negative}} \geq \underbrace{2h^* - h^{\text{firm}}}_{\text{worker turns negative}} \qquad (\text{since } h^{\text{soc}} \geq h^*).$$

Equality holds at the no-tax limit $\tau = 0$ (where $h^{\text{soc}} = h^*$ in this calibration); generically the joint threshold is strictly above the worker threshold. At baseline $(2h^{\text{soc}} - h^{\text{firm}},\, 2h^* - h^{\text{firm}}) = (0.30,\, 0.10)$, so for $\bar{h} \in (0.10, 0.30)$ the cap is welfare-reducing on aggregate (firm losses dominate) but the worker individually still benefits. The §3.4 "Reading" restates this distinction.

A.6 Heterogeneous-worker threshold: derivation of (2)

Replace α by the type-specific α_i in (7):

$$\Delta V_w^i(\bar{h}) = (\bar{h} - h^{\text{firm}})\Big[(1-\tau)w - \tfrac{\alpha_i}{2}(\bar{h} + h^{\text{firm}})\Big].$$

Since $\bar{h} < h^{\text{firm}}$, the leading factor is negative; therefore $\Delta V_w^i > 0$ iff the bracket is also negative:

$$(1-\tau)w < \tfrac{\alpha_i}{2}(\bar{h} + h^{\text{firm}}) \iff \alpha_i > \frac{2(1-\tau)w}{\bar{h} + h^{\text{firm}}} \equiv \alpha^*(\bar{h}),$$

which is (2). At $\bar{h} = h^{\text{soc}} = 0.40$: $\alpha^* = 2(0.6)(1)/(0.4 + 0.5) = 4/3 \approx 1.333$.

A.7 Aggregate welfare condition

With type distribution F and mean $\bar{\alpha} \equiv \int \alpha\, dF(\alpha)$, integrate (7) (with $\alpha \mapsto \alpha_i$) and add (5):

$$\overline{\Delta W}(\bar{h}) \equiv \int \Delta V_w^i(\bar{h})\, dF(\alpha) + \Delta V_f(\bar{h}) = (\bar{h} - h^{\text{firm}})\Big[(1-\tau)w - \tfrac{\bar{\alpha}}{2}(\bar{h} + h^{\text{firm}})\Big] - \tfrac{\beta}{2}(h^{\text{firm}} - \bar{h})^2.$$

Completing the square in $\bar{h}$ shows $\overline{\Delta W}(\bar{h}) > 0$ iff $\bar{h} \in (2\hat{h}^{\text{soc}} - h^{\text{firm}},\, h^{\text{firm}})$, where $\hat{h}^{\text{soc}} \equiv (A - \tau w)/(\bar{\alpha} + \beta)$ is the social optimum at the population-mean disutility $\bar{\alpha}$. The band has the same structure as (1) with $\alpha \mapsto \bar{\alpha}$. At baseline ($\bar{\alpha} = 2$, $\alpha^* = 1.333$ at $\bar{h} = h^{\text{soc}}$) the cap is welfare-improving on average; a workforce with mean $\bar{\alpha} = 1$ would have $\hat{h}^{\text{soc}} = (A - \tau w)/(\bar{\alpha} + \beta) = 1.6/3 \approx 0.53 > h^{\text{firm}}$, so any binding cap reduces aggregate welfare for that workforce.

B Formal derivation of the Q5 RTM bargain

This appendix derives the right-to-manage Nash-bargain equilibrium of §4.3 from primitives via Lagrangian and FOC. It supplies (i) the closed-form bargain equation behind the curves $w_j(\eta), h_j(\eta)$, (ii) the boundary conditions at $\eta = 0$ and $\eta \to 1$, (iii) the comparative statics $\partial w/\partial\eta > 0$, $\partial h/\partial\eta < 0$, and (iv) the welfare-restorer claim. The derivation matches the equilibrium condition implemented in `research/sim_models_v2.py:rtm_x` (annotated quadratic in lines 234–239 of that file).

B.1 Composite payoffs along the labour-demand curve

Per sector $j \in \{\text{G}, \text{B}\}$, the firm responds on its labour-demand curve $h_j^d(w) = (A_j - w)/\beta$. Substituting into the primitives of §3:

$$\begin{aligned} V_w(w; A_j) &\equiv V_w(w, h_j^d(w)) = \frac{(1-\tau)w(A_j - w)}{\beta} - \frac{\alpha(A_j - w)^2}{2\beta^2}, \\ V_f(w; A_j) &\equiv V_f(w, h_j^d(w); A_j) = \frac{(A_j - w)^2}{2\beta}, \end{aligned}$$

where V_f collapses to the standard Cournot-residual form $(A_j - w)^2/(2\beta)$ because the firm is on its own FOC. Reparameterise $x_j \equiv A_j - w = \beta h_j^d(w)$; then

$$V_w(x_j) = A_v\, x_j - B_v\, x_j^2, \qquad A_v \equiv \frac{(1-\tau)A_j}{\beta}, \quad B_v \equiv \frac{1-\tau}{\beta} + \frac{\alpha}{2\beta^2}; \qquad V_f(x_j) = \frac{x_j^2}{2\beta}. \tag{9}$$

B.2 Lagrangian for the generalised Nash programme

Union and firm choose w to maximise the asymmetric Nash product subject to the participation (IR) constraints:

$$\max_w \left[V_w(w; A_j) - V_w^0\right]^\eta V_f(w; A_j)^{1-\eta} \qquad \text{s.t.} \quad V_w \geq V_w^0,\ V_f \geq 0.$$

Taking logs (both factors strictly positive on the interior), the Lagrangian with multipliers $\mu_w, \mu_f \geq 0$ is

$$\mathcal{L}(x_j; \mu_w, \mu_f) = \eta \ln\!\big(V_w(x_j) - V_w^0\big) + (1-\eta) \ln V_f(x_j) + \mu_w\big(V_w(x_j) - V_w^0\big) + \mu_f V_f(x_j),$$

using $w = A_j - x_j$ and $dw/dx_j = -1$ (so the FOC in x_j is the negative of the FOC in w, leaving the zero-locus unchanged). On the interior, $\mu_w = \mu_f = 0$ and the FOC is

$$\frac{\eta\, V_w'(x_j)}{V_w(x_j) - V_w^0} + \frac{(1-\eta)\, V_f'(x_j)}{V_f(x_j)} = 0, \qquad V_w' = A_v - 2B_v x_j,\ V_f' = x_j/\beta. \tag{10}$$

B.3 Reduction to a quadratic in x_j

Multiply (10) through by $\big(V_w - V_w^0\big)V_f$, substitute $V_f = x_j^2/(2\beta)$, and cancel one factor of $x_j/(2\beta)$ (assuming $x_j \neq 0$, which excludes the no-employment corner $w = A_j$):

$$\eta\,(A_v - 2B_v x_j)\, x_j + 2(1-\eta)\,\big(A_v x_j - B_v x_j^2 - V_w^0\big) = 0.$$

Collecting terms:

$$2B_v\, x_j^2 - A_v(2-\eta)\, x_j + 2(1-\eta)\, V_w^0 = 0. \tag{11}$$

The equilibrium $x_j^\star(\eta)$ is the larger root (the smaller root is the IR-binding fixed point at $\eta = 0$ corner — see below), and the bargained wage and hours follow as $w_j(\eta) = A_j - x_j^\star(\eta)$, $h_j(\eta) = x_j^\star(\eta)/\beta$. This is exactly the equation solved by `rtm_x`; at calibration $(\alpha, \beta, \tau, A, V_w^0) = (2, 2, 0.4, 2, 0.05)$ it reads $1.1x^2 - 0.6(2-\eta)x + 0.1(1-\eta) = 0$, with $x^\star(0) = 1$ recovering $w = 1$, $h = 0.5$, and $x^\star(0.5) \approx 0.846$ giving $w \approx 1.154$, $h \approx 0.423$ (Figure 3 numbers).

B.4 Boundary conditions

(i) $\eta = 0$**.** Equation (11) becomes $2B_v x_j^2 - 2A_v x_j + 2V_w^0 = 0$, i.e. $V_w(x_j) = V_w^0$. The worker's IR binds and the firm picks the wage that strips all rent above the outside option; on the firm's labour-demand curve this delivers $h_j = h_j^{\text{firm}}$.

(ii) $\eta \to 1$**.** (11) becomes $2B_v x_j^2 - A_v x_j = 0$, with non-trivial root

$$x_j^\star(1) = \frac{A_v}{2B_v} = \frac{(1-\tau)A_j\,\beta}{2\beta(1-\tau) + \alpha} \implies h_j(1) = \frac{x_j^\star(1)}{\beta} = \frac{(1-\tau)A_j}{2\beta(1-\tau) + \alpha}.$$

SOC at the limit. Evaluating the bracket $4B_v x_j - A_v(2-\eta)$ at $(x, \eta) = (A_v/(2B_v), 1)$ gives $4B_v \cdot A_v/(2B_v) - A_v = 2A_v - A_v = A_v > 0$, so the second-order condition for the Nash maximum holds at the limit and $x_j^\star(1)$ is indeed the maximiser, not a corner. **Note on the text's $\eta \to 1$ statement.** §4.3 reads "$h_j \to h_j^* = (1-\tau)w_j/\alpha$" as $\eta \to 1$. The exact RTM limit gives $h_j(1)/h^*(w_j(1)) = \alpha/[\alpha + \beta(1-\tau)] < 1$ (equal to 0.625 at calibration). The reason: under RTM the firm *always* chooses on its own labour demand $h^d(w)$, so the worker's preferred hours $h^*(w) = (1-\tau)w/\alpha$ are reached only at the wage $w^{**} \equiv \alpha A_j/[\alpha + \beta(1-\tau)]$ where the two schedules intersect. The union, optimising on the demand curve as $\eta \to 1$, picks a wage *above* w^{**}, so $h^d(w_{\eta=1}) < h^*(w_{\eta=1})$. The "welfare-restorer" claim of §4.3 is unaffected — $h_j(\eta)$ still falls monotonically through h^{soc} — but the equality $h_j(1) = h_j^*$ should be replaced by the strict inequality $h_j(1) < h^*(w_j(1))$.

B.5 Walrasian benchmark and the §4.2 comparative statics

Maintained regularity. Two conditions on the outside option V_w^0 are imposed throughout. *(R1)* $A_v^2 > 4B_v V_w^0$ (strict discriminant), so the IR-binding equation $V_w(x_j) = V_w^0$ admits two distinct real roots. At calibration $A_v^2 = 0.36 > 4B_v V_w^0 = 4(0.55)(0.05) = 0.11$. *(R2)* $x_j^{\mathrm{FM}}(A_j)^2 > 2\beta^2 V_w^0/\alpha$ over the range of A_j considered, which delivers $\partial w_j^{\mathrm{FM}}/\partial A_j > 0$ (§B.5 below). At calibration $x_j^{\mathrm{FM}} = 1$ and $2\beta^2 V_w^0/\alpha = 0.2$, so (R2) holds with slack; on the sectoral range $A_j \in [1.8, 2.2]$ used in Q5 it remains satisfied (smallest value $x_b^{\mathrm{FM}} = 0.878$ gives $x^2 = 0.771 > 0.2$).

The competitive (free-market) outcome of §4.2 is the $\eta = 0$ specialisation of (11): $V_w(x_j) = V_w^0$, with larger root

$$x_j^{\mathrm{FM}}(A_j) = \frac{A_v + \sqrt{A_v^2 - 4B_v V_w^0}}{2B_v}, \qquad h_j^{\mathrm{FM}} = \frac{x_j^{\mathrm{FM}}}{\beta}, \qquad w_j^{\mathrm{FM}} = A_j - x_j^{\mathrm{FM}}. \tag{12}$$

Existence of the real larger root uses (R1). Differentiating $V_w(x_j; A_j) = V_w^0$ implicitly and using $\partial A_v/\partial A_j = (1-\tau)/\beta$:

$$\frac{\partial x_j^{\mathrm{FM}}}{\partial A_j} = \frac{(1-\tau)\, x_j^{\mathrm{FM}}}{\beta(2B_v x_j^{\mathrm{FM}} - A_v)} > 0,$$

positive because the larger root sits past the vertex $x = A_v/(2B_v)$ of the parabola V_w, where $V_w' = A_v - 2B_v x < 0$. Hence $\partial h_j^{\mathrm{FM}}/\partial A_j = \beta^{-1}\partial x_j^{\mathrm{FM}}/\partial A_j > 0$. For the wage, substituting $A_v x_j = B_v x_j^2 + V_w^0$ into the numerator above gives $\partial x_j^{\mathrm{FM}}/\partial A_j = (1-\tau)x_j^2/[\beta(B_v x_j^2 - V_w^0)] < 1$ whenever $x_j^2 > 2\beta^2 V_w^0/\alpha$ (using $\beta B_v - (1-\tau) = \alpha/(2\beta)$); this is exactly condition (R2), so $\partial w_j^{\mathrm{FM}}/\partial A_j = 1 - \partial x_j^{\mathrm{FM}}/\partial A_j > 0$ holds under the maintained regularity (and is verified at calibration). Together, these give the headline §4.2 comparative statics

$$\frac{\partial h_j^{\mathrm{FM}}}{\partial A_j} > 0, \qquad \frac{\partial w_j^{\mathrm{FM}}}{\partial A_j} > 0,$$

i.e. "hours follow productivity" under perfect labour mobility.

B.6 Comparative statics

Differentiating (11) implicitly in η:

$$\big[4B_v x_j - A_v(2-\eta)\big]\frac{dx_j}{d\eta} = -A_v x_j - 2V_w^0.$$

Bracket positivity (global on $[0,1)$**).** The larger root of (11) is $x_j^\star(\eta) = [A_v(2-\eta) + \sqrt{D(\eta)}]/(4B_v)$ with $D(\eta) \equiv A_v^2(2-\eta)^2 - 16B_v(1-\eta)V_w^0$. At $\eta = 0$, $D(0) = 4A_v^2 - 16B_v V_w^0 = 4(A_v^2 - 4B_v V_w^0) > 0$ by (R1); at $\eta \to 1$, $D(1) = A_v^2 > 0$; and D is continuous, so $D(\eta) > 0$ on $[0,1)$ with at most one interior dip — direct check of the discriminant of D in η confirms $D(\eta) > 0$ throughout.[8] Therefore $4B_v x_j^\star(\eta) = A_v(2-\eta) + \sqrt{D(\eta)} > A_v(2-\eta)$ on $[0,1)$, so the bracket $4B_v x_j - A_v(2-\eta) = \sqrt{D(\eta)} > 0$ strictly. The bracket equals $-V_w'(x_j^\star)\cdot$(positive) at the larger root, where $V_w'(x_j^\star) < 0$ (root past the vertex), so positivity of the bracket is the second-order condition for the Nash maximum.

The right-hand side is unambiguously negative for $x_j > 0$, $V_w^0 \geq 0$. Hence

$$\frac{dx_j}{d\eta} < 0 \implies \frac{dw_j}{d\eta} = -\frac{dx_j}{d\eta} > 0, \qquad \frac{dh_j}{d\eta} = \frac{1}{\beta}\frac{dx_j}{d\eta} < 0,$$

the union-pulls-wages-up / hours-down statics of §4.3.

B.7 The welfare-restorer claim

Joint surplus. W is concave in h with peak at $h_j^{\mathrm{soc}} \in (h_j^*, h_j^{\mathrm{firm}})$ (§3, ordering at the firm-power baseline). Since $h_j(\eta)$ is continuous and strictly decreasing in η with $h_j(0) = h_j^{\mathrm{firm}}$, by the intermediate-value theorem there exists a unique $\hat{\eta}_j \in (0,1)$ with $h_j(\hat{\eta}_j) = h_j^{\mathrm{soc}}$. Combining with the quadratic identity (6) of Appendix A,

$$W\big(h_j(\eta)\big) - W(h_j^{\mathrm{firm}}) = -\tfrac{\alpha+\beta}{2}\Big[(h_j^{\mathrm{soc}} - h_j(\eta))^2 - (h_j^{\mathrm{soc}} - h_j^{\mathrm{firm}})^2\Big],$$

which is increasing on $[0, \hat{\eta}_j]$ (the union pulls hours *toward* the social bliss) and decreasing on $[\hat{\eta}_j, 1]$ (overshoot region).

[8] D is itself a quadratic in η with positive leading coefficient A_v^2 and minimum value $A_v^2 - 16B_v V_w^0/(\text{constant})$; under (R1) the minimum exceeds zero on $[0,1)$. The check is mechanical.

Worker surplus is monotone in η. On the labour-demand curve $V_w(x_j) = A_v x_j - B_v x_j^2$ is a concave parabola with maximum at $\hat{x} \equiv A_v/(2B_v) = x_j^\star(1)$. At $\eta = 0$, $V_w(x_j^\star(0)) = V_w^0$, so $x_j^\star(0)$ is the larger root of $V_w(x) = V_w^0$:

$$x_j^\star(0) = \frac{A_v + \sqrt{A_v^2 - 4B_v V_w^0}}{2B_v} > \frac{A_v}{2B_v} = \hat{x} = x_j^\star(1),$$

which together with monotonicity of $x_j^\star(\eta)$ (§B.5) gives $x_j^\star(\eta) \in (\hat{x}, x_j^\star(0)]$ for all $\eta \in [0,1)$. On this interval $V_w'(x) = A_v - 2B_v x < 0$, so as $x_j^\star$ falls in η, V_w rises. Formally,

$$\frac{dV_w}{d\eta} = V_w'(x_j^\star) \cdot \frac{dx_j^\star}{d\eta} > 0 \qquad \text{for all } \eta \in [0,1),$$

a product of two negatives.

Conclusion. Joint surplus is hump-shaped in η on $[0,1]$ with peak at $\hat{\eta}_j$, and worker surplus is strictly increasing in η. A planner with any positive worker weight $\lambda > 0$ in the Q7 weighting $W^\lambda(\eta) = (1+\lambda)V_w + (1-\lambda)V_f$ thus strictly prefers $\eta > 0$ to $\eta = 0$, since V_f is bounded above by its $\eta = 0$ value while V_w is strictly higher.